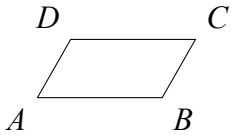
1 Vectors I

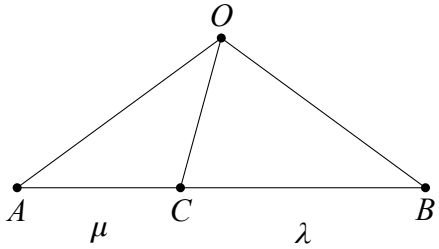
DRAFT: Working draft. All examples are taken from your Vectors Summary Notes; the workings are mine, checked with sympy.

1.1 Basic Formulas

	Formula	Note
Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{x^2+y^2+z^2}$	
Unit vector	$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$	$\lvert\hat{\mathbf{a}}\rvert = 1$
Parallel vectors	$\mathbf{a} = k\mathbf{b}$ $\quad$ $\mathbf{a}\times\mathbf{b} = \mathbf{0}$	k scalar constant
Collinear A, B, C	$\overrightarrow{AB} \parallel \overrightarrow{BC}$ (or any two of $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{AC}$)	Try drawing the diagram and take any two vectors, to show that they are parallel.
Parallelogram $ABCD$	$\overrightarrow{AB} = \overrightarrow{DC}$, $\quad$ $\overrightarrow{AD} = \overrightarrow{BC}$	

The Ratio Theorem

Let the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, and suppose that $\lambda AC = \mu CB$. Then the position vector of C is given by the formula

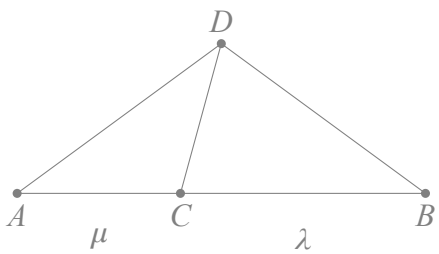


Ratio Theorem

$$\mathbf{c} = \frac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$$

$AC : CB = \mu : \lambda$, so C divides AB with μ beside A and λ beside B.

The same theorem, starting from a point D instead of O:

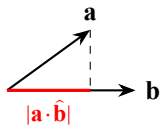
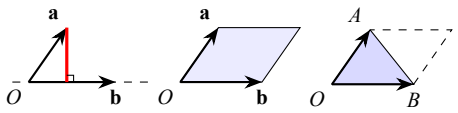


Ratio Theorem

$$\overrightarrow{DC} = \frac{\lambda\overrightarrow{DA} + \mu\overrightarrow{DB}}{\lambda + \mu}$$

1.2 Scalar Product and Vector Product

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}, \qquad \mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}, \qquad \theta=\text{non-reflex } \angle AOB$$

	Scalar Product (Dot)	Vector Product (Cross)
Definition	$\mathbf{a}\cdot\mathbf{b}=a_1b_1+a_2b_2+a_3b_3$	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix} a_2b_3-a_3b_2 \\ -(a_1b_3-a_3b_1) \\ a_1b_2-a_2b_1 \end{pmatrix}$
Result	scalar	vector, $\perp$ to both $\mathbf{a}$ and $\mathbf{b}$
Formula	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$
Angle	$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$	$\sin\theta=\dfrac{\lvert\mathbf{a}\times\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ this method only finds the basic angle possible, but usually undesirable
Properties	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$ $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$ $\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$ $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$	$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$ $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$ $\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$ $\mathbf{a}\times\mathbf{a}=\mathbf{0}$
Test	$\mathbf{a}\cdot\mathbf{b}=0 \iff \mathbf{a}\perp\mathbf{b}$ >0 acute, <0 obtuse, $=0$ right angle	$\mathbf{a}\times\mathbf{b}=\mathbf{0} \iff \mathbf{a}\parallel\mathbf{b}$ (nonzero $\mathbf{a}$, $\mathbf{b}$)
Applications	Length of projection $=\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ Projection vector $=(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$	Perpendicular distance $=\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ Area of $\triangle OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ Area of parallelogram $=\lvert\mathbf{a}\times\mathbf{b}\rvert$
Geometrical meaning	 $\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ = length of projection of $\mathbf{a}$ onto $\mathbf{b}$	 perpendicular distance from A to line along $\mathbf{b}$ $\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ area of parallelogram, sides $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$ 2× area of $\triangle OAB$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$
Note	$\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law	no cancellation law $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is a vector, but triple cross products are usually not tested

Note

Projection of $\mathbf{a}$ onto $\mathbf{b}$ $\neq$ projection of $\mathbf{b}$ onto $\mathbf{a}$. Perpendicular distance from A to OB $\neq$ from B to OA. Triangles without O as a vertex: $\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$.

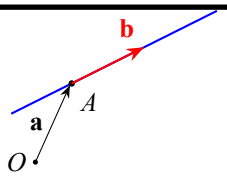
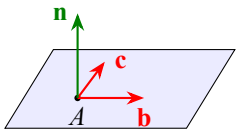
Useful identity

For any vectors $\mathbf{a}$, $\mathbf{b}$: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

$\because \mathbf{a}\times\mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$.

2 Vectors II

2.1 Equations of Lines and Planes

	Vector	Cartesian	What you need
Line	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $(\mathbf{r}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ (same line: $\mathbf{r}-\mathbf{a}$ // $\mathbf{b}$)	$\frac{x-a}{d}=\frac{y-b}{e}=\frac{z-c}{f}$ $\mathbf{b}=\begin{pmatrix}d\\e\\f\end{pmatrix}$; if $d=0$: $x=a$ separately all coefficients of x, y and z in the numerators must be 1	**Need:** 1 point + 1 direction or 2 points A, B: $\mathbf{b}=\overrightarrow{AB}$
Plane	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ $\mathbf{r}\cdot\mathbf{n}=d$ (scalar product form)	$ax+by+cz=d$	**Need:** 1 point + 2 non-parallel directions // plane or 1 point + normal $\mathbf{n}$ $\mathbf{n}=\mathbf{b}\times\mathbf{c},\quad d=\mathbf{a}\cdot\mathbf{n}$ 3 points: $\mathbf{b}=\overrightarrow{AB}$, $\mathbf{c}=\overrightarrow{AC}$; contains a line: use its direction

$\mathbf{a}$: position vector of any point on the line/plane. $\mathbf{b}$, $\mathbf{c}$: any two linearly independent directions parallel to the plane.

The equation of any line is not unique.

Note

To check if a given point A lies on

- a line: $\mathbf{a}_A=\mathbf{a}+\lambda\mathbf{b}$ (one λ satisfies all 3 components)
- a plane: $\mathbf{a}_A\cdot\mathbf{n}=d$

Example 2.1

Line through $(1,2,3)$, $(3,2,2)$

$$\mathbf{b}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$\frac{x-1}{2}=\frac{z-3}{-1},\quad y=2$$

Example 2.2

Plane through $B(0,1,1)$, $C(1,1,3)$, $D(2,0,2)$

$$\mathbf{n}=\overrightarrow{BC}\times\overrightarrow{BD}=\begin{pmatrix}1\\0\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\3\\-1\end{pmatrix}$$

$$d=\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\-1\end{pmatrix}=2$$

$$2x+3y-z=2$$

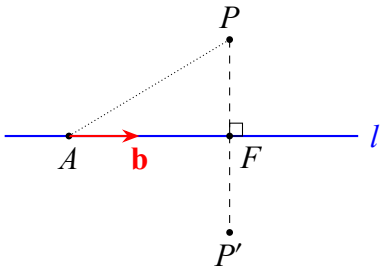
2.2 Interactions Map

	Point	Line	Plane
Point	distance $\lvert\overrightarrow{AB}\rvert$		
Line	• on line? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / intersect / skew §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in line §2.7	
Plane	• on plane? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / lies in / intersect §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in plane §2.7	• // / same / intersect §2.4 • line of intersection §2.4 • angle §2.6 • distance (//) §2.8

Always draw the diagram first.

2.3 Foot / Distance / Reflection

Point P – Line $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$

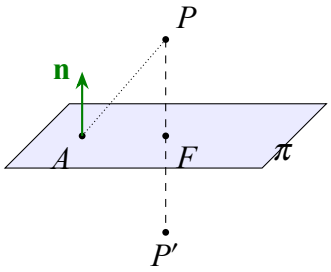


1. $\overrightarrow{OF}=\mathbf{a}+\lambda\mathbf{b}$ for some $\lambda\in\mathbb{R}$
2. $\overrightarrow{PF}\cdot\mathbf{b}=0 \Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

$$PF=\lvert\overrightarrow{AP}\times\hat{\mathbf{b}}\rvert$$

Point P – Plane $\mathbf{r}\cdot\mathbf{n}=d$



1. $l_{PF}:\ \mathbf{r}=\mathbf{p}+\lambda\mathbf{n},\ \lambda\in\mathbb{R}$
2. $(\mathbf{p}+\lambda\mathbf{n})\cdot\mathbf{n}=d$ (intersection between line & plane)
   $\Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

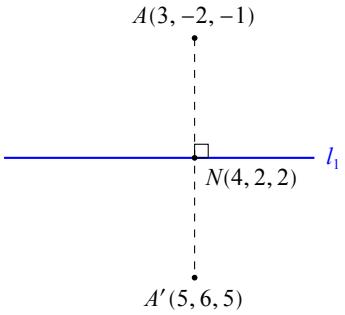
$$PF=\lvert\overrightarrow{AP}\cdot\hat{\mathbf{n}}\rvert$$

Reflection (and foot of perpendicular) of point to line

Example 2.3

Find the foot of the perpendicular N from $A(3,-2,-1)$ to the line l_1, and the reflection A' of A in l_1, where

$$l_1: \mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}, \quad \lambda\in\mathbb{R}$$



$$\overrightarrow{AN}=\begin{pmatrix}-2+3\lambda\\4\\4-\lambda\end{pmatrix}$$

$$\overrightarrow{AN}\cdot\begin{pmatrix}3\\0\\-1\end{pmatrix}=0$$

$$10\lambda-10=0 \Rightarrow \lambda=1$$

$$\therefore N=(4,2,2)$$

$$\overrightarrow{OA'}=2\begin{pmatrix}4\\2\\2\end{pmatrix}-\begin{pmatrix}3\\-2\\-1\end{pmatrix}$$

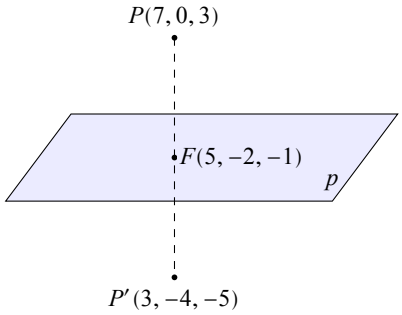
$$\therefore A'=(5,6,5)$$

Reflection (and foot of perpendicular) of point to plane

Example 2.4

Find the foot of the perpendicular F from $P(7,0,3)$ to the plane p, and the reflection P' of P in p, where

$$p: \mathbf{r}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$



$$\left(\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$

$$13+6\lambda=1 \Rightarrow \lambda=-2$$

$$\therefore F=(5,-2,-1)$$

$$\overrightarrow{OP'}=2\begin{pmatrix}5\\-2\\-1\end{pmatrix}-\begin{pmatrix}7\\0\\3\end{pmatrix}$$

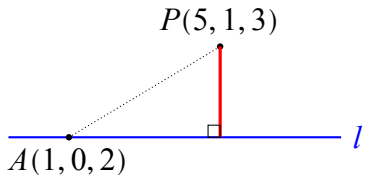
$$\therefore P'=(3,-4,-5)$$

Shortest distance from point to line

Example 2.5

Find the shortest distance from $P(5,1,3)$ to the line

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$



$$\text{shortest distance} = |\overrightarrow{AP} \times \hat{\mathbf{b}}|$$

$$= \left| \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \times \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right|$$

$$= \left| \frac{1}{3} \begin{pmatrix} 3 \\ -6 \\ -6 \end{pmatrix} \right| = \left| \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} \right|$$

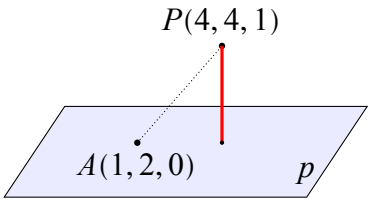
$$= 3$$

Shortest distance from point to plane

Example 2.6

Find the shortest distance from $P(4,4,1)$ to the plane

$$p: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 4$$



$$\text{shortest distance} = |\overrightarrow{AP} \cdot \hat{\mathbf{n}}|$$

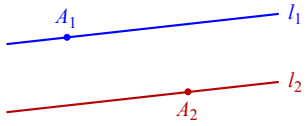
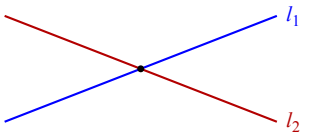
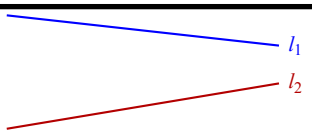
$$= \left| \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right|$$

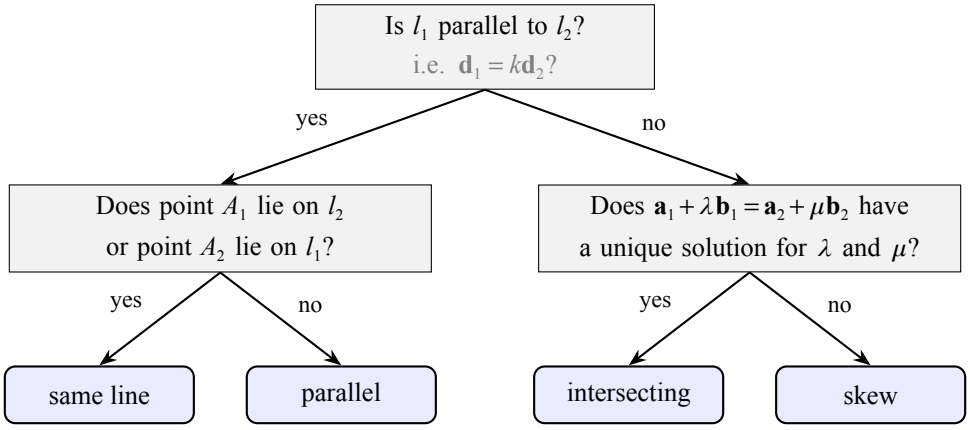
$$= \left| \frac{1}{3}(6+2-2) \right|$$

$$= 2$$

2.4 Relationships and Intersections

Line – Line $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{d}_1$, $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{d}_2$

	Parallel	Intersecting	Skew
Diagram	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$
Condition	$\mathbf{d}_1 \parallel \mathbf{d}_2$ and $A_1 \notin l_2$, $A_2 \notin l_1$	$\mathbf{d}_1 \nparallel \mathbf{d}_2$, unique solution for λ and μ	$\mathbf{d}_1 \nparallel \mathbf{d}_2$, no solution for λ and μ



Note

Definition of **skew lines**:

- non parallel
- non intersecting

Thus, to prove that two lines are skew, we must first state that $\mathbf{d}_1 \neq k\mathbf{d}_2$, so l_1 and l_2 are not parallel, before proceeding with simultaneous equations to show they have no solution.

Example 2.7

Determine whether l_1 and l_2 are intersecting or skew lines.

$$l_1: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$l_2: \mathbf{r}=\begin{pmatrix}2\\12\\5\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\0\end{pmatrix},\quad \mu\in\mathbb{R}$$

$$1+\lambda=2-\mu \qquad —(1)$$

$$2\lambda=12+3\mu \qquad —(2)$$

$$2+\lambda=5 \qquad —(3)$$

From (1) and (3), $\lambda=3,\quad \mu=-2$

Sub into (2), LHS $=6,\quad$ RHS $=12+3(-2)=6$

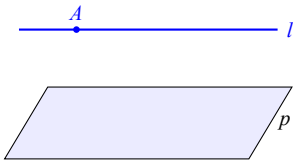
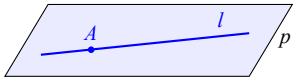
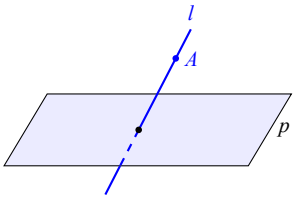
$\therefore$ LHS $=$ RHS

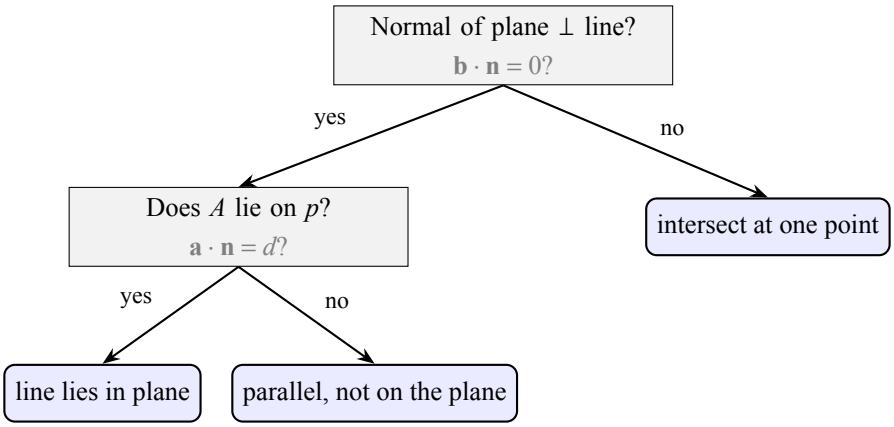
$$\text{Sub } \lambda=3 \text{ into } l_1: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+3\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}4\\6\\5\end{pmatrix}$$

They intersect at $(4, 6, 5)$

Solve 2 of the 3 equations, sub into the 3rd: LHS = RHS ⇒ intersect; LHS ≠ RHS ⇒ skew.

Line – Plane $\quad l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \pi: \mathbf{r}\cdot\mathbf{n}=d$

	Parallel, not on the plane	Line lies in the plane	Intersects at one point
Diagram	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$
Condition	• normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A does not lie on p: $\mathbf{a}\cdot\mathbf{n}\neq d$	• normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A lies on p: $\mathbf{a}\cdot\mathbf{n}=d$	• normal not $\perp$ line: $\mathbf{b}\cdot\mathbf{n}\neq 0$



Example 2.8

Find the point of intersection between the line l and plane p.

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$p: \mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 5$$

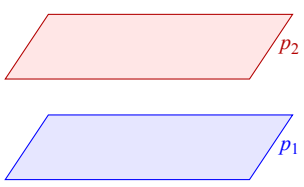
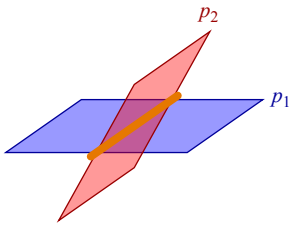
$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+\lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

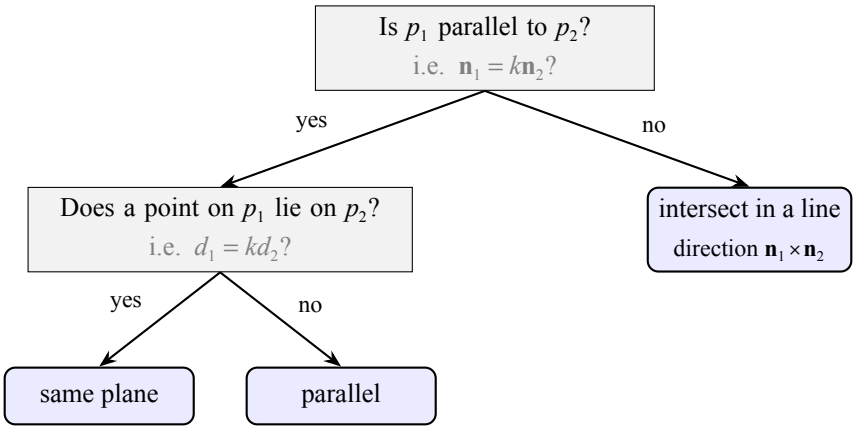
$$\begin{pmatrix} 1+\lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 5$$

$$8 - 2\lambda = 5 \Rightarrow \lambda = \tfrac{3}{2}$$

$$\therefore (2.5, 0, 2.5)$$

Plane – Plane $p_1 : \mathbf{r} \cdot \mathbf{n}_1 = d_1,\ p_2 : \mathbf{r} \cdot \mathbf{n}_2 = d_2$

	Parallel	Intersecting in a line
Diagram		
Condition	$\mathbf{n}_1 \parallel \mathbf{n}_2, p_1 \neq p_2$	$\mathbf{n}_1 \nparallel \mathbf{n}_2$



Example 2.9

Find the intersection between the planes:

$$3x+2y+z=10$$

$$x-2y+z=5$$

$$\text{GC: } x=\frac{15}{4}-\frac{1}{2}z$$

$$y=-\frac{5}{8}+\frac{1}{4}z$$

$$z=z$$

$$\mathbf{r}=\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}\frac{15}{4}-\frac{1}{2}z\\-\frac{5}{8}+\frac{1}{4}z\\z\end{pmatrix}$$

$$=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+z\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix}$$

$$\therefore l:\ \mathbf{r}=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+z\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix},\quad z\in\mathbb{R}$$

$\mathbf{d}=\mathbf{n}_1\times\mathbf{n}_2=(4,-2,-8)^{\mathrm{T}}\,//\,(2,-1,-4)^{\mathrm{T}}$ checks the direction.

Example 2.10

Sometimes, we may not be able to use our GC to find the line of intersection between two planes. The steps to find it manually are:

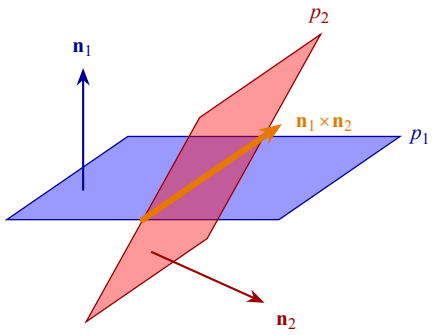
$$p_1: \mathbf{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=6, \qquad p_2: \mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=2$$

Step 1: direction $=\mathbf{n}_1\times\mathbf{n}_2$

$$\mathbf{d}=\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$=\begin{pmatrix}1(2)-1(-1)\\1(1)-1(2)\\1(-1)-1(1)\end{pmatrix}$$

$$=\begin{pmatrix}3\\-1\\-2\end{pmatrix}$$



Step 2: find a point that lies on both planes

$$p_1: x+y+z=6$$

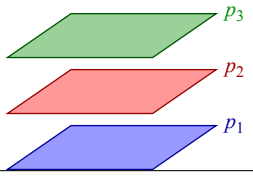
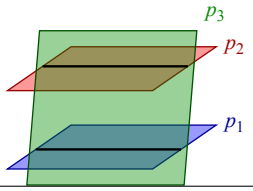
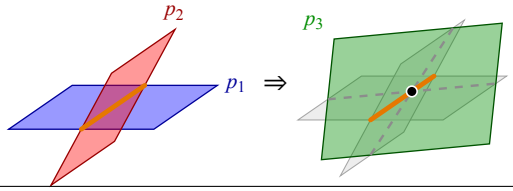
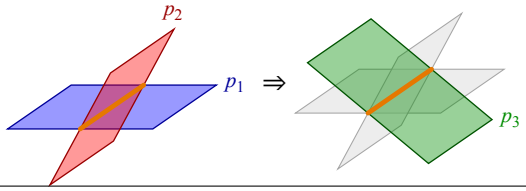
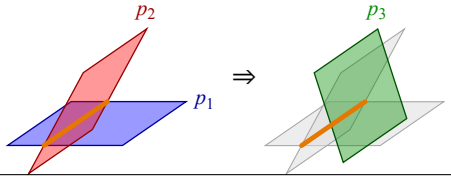
$$p_2: x-y+2z=2$$

$$\text{Let } z=0$$

$$\left.\begin{aligned}x+y&=6\\x-y&=2\end{aligned}\right\}\Rightarrow\begin{aligned}x&=4\\y&=2\end{aligned}$$

$$\therefore l: \mathbf{r}=\begin{pmatrix}4\\2\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\-1\\-2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

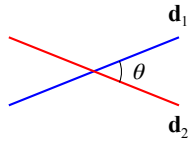
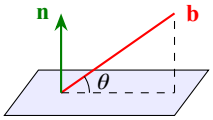
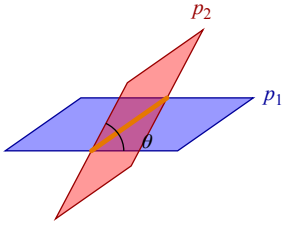
2.5 Interaction between 3 Planes

Case	Diagram(s)	Condition
1. All parallel		—
2. Two parallel, one not		—
3. All intersect at 1 point (most common)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ intersects p_3 (see p. 8)
4. All intersect in a line		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ lies on p_3 (see p. 8)
5. Form a triangular prism (no single point of intersection)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ does not lie on p_3 (see p. 8)

Note

To analyse the interaction between 3 arbitrary planes, we can study how the **line of intersection between p_1 and p_2** interacts with p_3.

2.6 Angles

Line – Line	Line – Plane	Plane – Plane
		
$\cos\theta = \dfrac{\lvert\mathbf{d}_1\cdot\mathbf{d}_2\rvert}{\lvert\mathbf{d}_1\rvert\lvert\mathbf{d}_2\rvert}$	$\sin\theta = \left\lvert\dfrac{\mathbf{n}\cdot\mathbf{b}}{\lvert\mathbf{n}\rvert\lvert\mathbf{b}\rvert}\right\rvert$	$\cos\theta = \left\lvert\dfrac{\mathbf{n}_1\cdot\mathbf{n}_2}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}\right\rvert$
directions	sin: $\angle(\mathbf{n},\mathbf{b}) = 90^\circ - \theta$	normals

θ = acute angle.

Each formula has a modulus in the numerator $\because \theta$ must be acute.

Example 2.11

Find the angle between the planes $3x-y+2z=3$ and $y=2$.

$$\cos\theta = \frac{|-1|}{\sqrt{14}\cdot 1}$$
$$\theta = 74.5^\circ$$

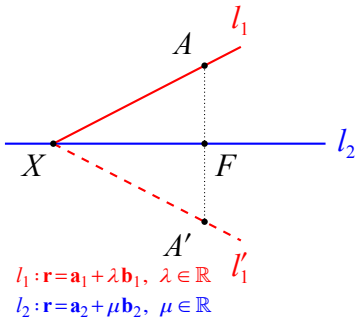
Example 2.12

Find the angle between the line l and plane p.

$$l:\ \mathbf{r} = \begin{pmatrix}-2\\-4\\5\end{pmatrix} + \lambda\begin{pmatrix}3\\5\\-4\end{pmatrix},\quad \lambda\in\mathbb{R}$$
$$p:\ \mathbf{r}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix} = 1$$
$$\sin\theta = \frac{5}{\sqrt{50}\cdot 1} = \frac{1}{\sqrt{2}}$$
$$\theta = 45^\circ$$

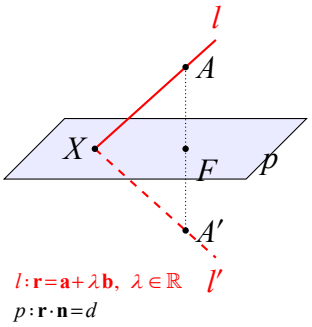
2.7 Reflection of a Line

Line l_1 in line l_2 (intersect at X)



1. $X=l_1\cap l_2$
2. $A\in l_1$: foot F on l_2
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l_1': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

Line l in plane p (intersect at X)



1. $X=l\cap p$
2. $A\in l$: foot F on p
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

If $l\ /\!/\ p$: l' has the same direction, through A'.

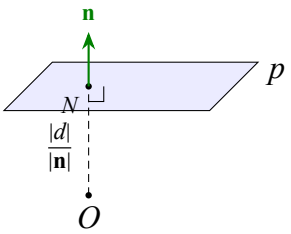
DRAFT: 4.7 Reflection (Line about Line) is listed in your source but its page is missing; the steps on the left are the standard method. Add an example from an official answer sheet?

2.8 Distance Between Parallel Planes

$p: \mathbf{r}\cdot\mathbf{n}=d$

Distance of plane from origin

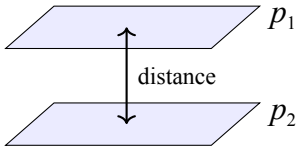
$$\text{distance from } O=\frac{|d|}{|\mathbf{n}|}$$



$p_1: \mathbf{r}\cdot\mathbf{n}=d_1,\qquad p_2: \mathbf{r}\cdot\mathbf{n}=d_2$

Distance

$$\text{distance}=\left|\frac{d_1}{|\mathbf{n}|}-\frac{d_2}{|\mathbf{n}|}\right| \qquad \text{plane to } O=\frac{|k|}{|\mathbf{n}|}$$



Same $\mathbf{n}$ for both planes first. $\mathbf{r}\cdot\mathbf{n}=0$: plane through O.

Example 2.13

Find the distance between the two planes $x+2y-3z=2$ and $2x+4y-6z=-10$.

$$2x+4y-6z=-10 \Rightarrow x+2y-3z=-5$$

$$\text{distance}=\left|\frac{2}{\sqrt{14}}-\frac{-5}{\sqrt{14}}\right|=\frac{7}{\sqrt{14}}$$